

Security requirements in IoT-enabled application services

IoT-enabled devices communicate through protocols like Message Queue Telemetry Transport (MQTT), which is an open messaging and lightweight protocol for IoT. This involves personal area networks (PANs) or wide area networks (WANs) connecting multiple data centres through public or private cloud services.

In this network, IoT security on infrastructure components and IoT application security on IoT-enabled edge services (like secure access service edge (SASE)) are important to protect connected devices and application data. Some of the core principles for developing a security strategy for IoT applications and infrastructure services are:

- Data encryption at rest to protect data when it is stored in the data storage or application storage facility. This could be through encryption or device level cipher protection to avoid tampering data access from outside.
- Data encryption in transit or in motion to protect data when transporting through a communication protocol. This could be through secured tunnel or safeguarding network services using secured protocols like SSL, TLS, FTPS, etc.
- Device identity to protect unauthorised devices being connected to IoT network of services. Each device in the IoT implementation is identified through unique device identity. This is used to authenticate (verify device access) and authorise (permit device connection) in the IoT ecosystem of connected devices.
- Protect device authentication through a token-based authentication and authorisation framework like OAuth 2.0.
- Policy management to provide role-based access control, along with user

principles and policy definitions to enable device protection facilities in the IoT ecosystem. This will enable access management for all the connected services.

- Two-way protection in all services through certificate based authentication like public key infrastructure (PKI) and organisation-based access control (OrBAC).

When using MQTT for machine-to-machine network communication, it's essential to ensure device authorisation and data protection. IoT security and IoT application security play critical roles in the IoT ecosystem.

Edge computing for IoT services

Edge computing for IoT services is a distributed computing paradigm that brings computation and data storage closer to data sources, such as IoT devices and sensors. This allows for real-time data processing and analysis, reduced latency, and improved bandwidth efficiency.

Edge computing is particularly well-suited for IoT applications because it can help to address some of the key challenges of IoT, such as:

- **Handling large volumes of data:** IoT devices generate large volumes of data, which can be costly and time-consuming to transmit to the cloud. Edge computing can help reduce this amount of data by processing and filtering it at the edge.
- **Reducing latency:** IoT applications often require real-time data processing and analysis in order to make timely decisions. Edge computing can help to reduce latency by bringing computation closer to the sources of data.
- **Improving reliability:** IoT devices are often deployed in remote or harsh environments where network connectivity may be unreliable. Edge computing can help to improve reliability by allowing IoT devices to continue operating even when they are not connected to the cloud.

Edge computing for IoT services can be used to implement a wide variety of applications, including:

- **Real-time monitoring and control:** Edge computing can be used to monitor and control IoT devices in real time. This can be used for applications such as predictive maintenance, traffic management, and industrial automation.
 - **Video analytics:** Edge computing can be used to perform video analytics on IoT devices, such as security cameras and traffic cameras. This can be used for applications like object detection, motion detection, and facial recognition.
 - **Machine learning:** Edge computing can be used to train and deploy machine learning models on IoT devices. This allows for real-time predictions and decisions to be made at the edge, without the need to send data to the cloud.
- Here are some examples of edge computing for IoT services in use today.
- **Smart cities:** Edge computing is being used to power smart city applications such as real-time traffic monitoring, parking management, and air quality monitoring.
 - **Industrial automation:** Edge computing is being used to improve the efficiency and reliability of industrial automation processes. For example, edge computing can be used to monitor and control machines in real time, and to perform predictive maintenance to prevent downtime.
 - **Retail:** Edge computing is being used to improve the customer experience in retail stores. For example, it can be used to power personalised recommendations, in-store navigation, and queue management systems.
- Overall, edge computing is a powerful technology that can be used to improve the performance, reliability, and security of IoT services.